SG No. 223 O_{h}^{3} $Pm\bar{3}n$ [cubic]

* plus set: $+[0,\,0,\,0]$
* Wyckoff site: `2a`, site symmetry: `m-3.`

Table 1: Wyckoff bond: `6a@2a`, bond symmetry: `mmm..`

No.	vector	center	mapping
1	$[X,\,0,\,0]$	$[0,\,0,\,0]$	`[1,-2,-3,4,-25,26,27,-28]`
2	$[0,\,X,\,0]$	$[0,\,0,\,0]$	`[5,-6,-7,8,-29,30,31,-32]`
3	$[0,\,0,\,X]$	$[0,\,0,\,0]$	`[9,-10,-11,12,-33,34,35,-36]`
4	$[0,\,X,\,0]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[13,-14,-15,16,-37,38,39,-40]`
5	$[X,\,0,\,0]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[17,-18,-19,20,-41,42,43,-44]`
6	$[0,\,0,\,-X]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[21,-22,-23,24,-45,46,47,-48]`

Table 2: Wyckoff bond: `8b@2a`, bond symmetry: `.-3.`

No.	vector	center	mapping
1	$[X,\,X,\,X]$	$[0,\,0,\,0]$	`[1,5,9,-25,-29,-33]`
2	$[-X,\,-X,\,X]$	$[0,\,0,\,0]$	`[2,7,12,-26,-31,-36]`
3	$[-X,\,X,\,-X]$	$[0,\,0,\,0]$	`[3,8,10,-27,-32,-34]`
4	$[X,\,-X,\,-X]$	$[0,\,0,\,0]$	`[4,6,11,-28,-30,-35]`
5	$[X,\,X,\,-X]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[13,17,21,-37,-41,-45]`
6	$[-X,\,-X,\,-X]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[14,19,24,-38,-43,-48]`
7	$[X,\,-X,\,X]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[15,20,22,-39,-44,-46]`
8	$[-X,\,X,\,X]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[16,18,23,-40,-42,-47]`

Table 3: Wyckoff bond: `12c@2a`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X,\,Y,\,0]$	$[0,\,0,\,0]$	`[1,-2,-25,26]`
2	$[-X,\,Y,\,0]$	$[0,\,0,\,0]$	`[3,-4,-27,28]`
3	$[0,\,X,\,Y]$	$[0,\,0,\,0]$	`[5,-6,-29,30]`
4	$[0,\,-X,\,Y]$	$[0,\,0,\,0]$	`[7,-8,-31,32]`
5	$[Y,\,0,\,X]$	$[0,\,0,\,0]$	`[9,-10,-33,34]`
6	$[Y,\,0,\,-X]$	$[0,\,0,\,0]$	`[11,-12,-35,36]`
7	$[Y,\,X,\,0]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[13,-14,-37,38]`
8	$[Y,\,-X,\,0]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[15,-16,-39,40]`
9	$[X,\,0,\,-Y]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[17,-18,-41,42]`
10	$[-X,\,0,\,-Y]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[19,-20,-43,44]`
11	$[0,\,Y,\,-X]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[21,-22,-45,46]`
12	$[0,\,Y,\,X]$	$\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$	`[23,-24,-47,48]`

Table 4: Wyckoff bond: `24d@2a`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\,Y,\,Z]$	$[0,\,0,\,0]$	`[1,-25]`

continued ...

Table 4

No.	vector	center	mapping
2	$[-X, -Y, Z]$	$[0, 0, 0]$	`[2,-26]`
3	$[-X, Y, -Z]$	$[0, 0, 0]$	`[3,-27]`
4	$[X, -Y, -Z]$	$[0, 0, 0]$	`[4,-28]`
5	$[Z, X, Y]$	$[0, 0, 0]$	`[5,-29]`
6	$[Z, -X, -Y]$	$[0, 0, 0]$	`[6,-30]`
7	$[-Z, -X, Y]$	$[0, 0, 0]$	`[7,-31]`
8	$[-Z, X, -Y]$	$[0, 0, 0]$	`[8,-32]`
9	$[Y, Z, X]$	$[0, 0, 0]$	`[9,-33]`
10	$[-Y, Z, -X]$	$[0, 0, 0]$	`[10,-34]`
11	$[Y, -Z, -X]$	$[0, 0, 0]$	`[11,-35]`
12	$[-Y, -Z, X]$	$[0, 0, 0]$	`[12,-36]`
13	$[Y, X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[13,-37]`
14	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[14,-38]`
15	$[Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[15,-39]`
16	$[-Y, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[16,-40]`
17	$[X, Z, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[17,-41]`
18	$[-X, Z, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[18,-42]`
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[19,-43]`
20	$[X, -Z, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[20,-44]`
21	$[Z, Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[21,-45]`
22	$[Z, -Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[22,-46]`
23	$[-Z, Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[23,-47]`
24	$[-Z, -Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[24,-48]`

* Wyckoff site: `6b`, site symmetry: `mmm..`

Table 5: Wyckoff bond: `6a@6b`, bond symmetry: `mmm..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2,-3,4,-25,26,27,-28]`
2	$[0, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-6,-7,8,-29,30,31,-32]`
3	$[0, 0, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-10,-11,12,-33,34,35,-36]`
4	$[0, X, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[13,-14,-15,16,-37,38,39,-40]`
5	$[X, 0, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[17,-18,-19,20,-41,42,43,-44]`
6	$[0, 0, -X]$	$\left[0, 0, \frac{1}{2}\right]$	`[21,-22,-23,24,-45,46,47,-48]`

Table 6: Wyckoff bond: `6b@6b`, bond symmetry: `mmm..`

No.	vector	center	mapping
1	$[0, 0, X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,-3,-4,-25,-26,27,28]`
2	$[X, 0, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,6,-7,-8,-29,-30,31,32]`
3	$[0, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10,-11,-12,-33,-34,35,36]`
4	$[0, 0, -X]$	$\left[0, \frac{1}{2}, 0\right]$	`[13,14,-15,-16,-37,-38,39,40]`
5	$[0, X, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[17,18,-19,-20,-41,-42,43,44]`
6	$[X, 0, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[21,22,-23,-24,-45,-46,47,48]`

Table 7: Wyckoff bond: `6c@6b`, bond symmetry: `mmm..`

No.	vector	center	mapping
1	$[0, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2,3,-4,-25,26,-27,28]`
2	$[0, 0, X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-6,7,-8,-29,30,-31,32]`
3	$[X, 0, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-10,11,-12,-33,34,-35,36]`
4	$[X, 0, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[13,-14,15,-16,-37,38,-39,40]`
5	$[0, 0, -X]$	$\left[\frac{1}{2}, 0, 0\right]$	`[17,-18,19,-20,-41,42,-43,44]`
6	$[0, X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[21,-22,23,-24,-45,46,-47,48]`

Table 8: Wyckoff bond: `12d@6b`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2,-25,26]`
2	$[-X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-4,-27,28]`
3	$[0, X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-6,-29,30]`
4	$[0, -X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[7,-8,-31,32]`
5	$[Y, 0, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-10,-33,34]`
6	$[Y, 0, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[11,-12,-35,36]`
7	$[Y, X, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[13,-14,-37,38]`
8	$[Y, -X, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[15,-16,-39,40]`
9	$[X, 0, -Y]$	$\left[\frac{1}{2}, 0, 0\right]$	`[17,-18,-41,42]`
10	$[-X, 0, -Y]$	$\left[\frac{1}{2}, 0, 0\right]$	`[19,-20,-43,44]`
11	$[0, Y, -X]$	$\left[0, 0, \frac{1}{2}\right]$	`[21,-22,-45,46]`
12	$[0, Y, X]$	$\left[0, 0, \frac{1}{2}\right]$	`[23,-24,-47,48]`

Table 9: Wyckoff bond: `12e@6b`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[Y, 0, X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-3,-25,27]`
2	$[-Y, 0, X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-4,-26,28]`
3	$[X, Y, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-7,-29,31]`
4	$[X, -Y, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[6,-8,-30,32]`
5	$[0, X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-11,-33,35]`
6	$[0, X, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[10,-12,-34,36]`
7	$[0, Y, -X]$	$\left[0, \frac{1}{2}, 0\right]$	`[13,-15,-37,39]`
8	$[0, -Y, -X]$	$\left[0, \frac{1}{2}, 0\right]$	`[14,-16,-38,40]`
9	$[Y, X, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[17,-19,-41,43]`
10	$[-Y, X, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[18,-20,-42,44]`
11	$[X, 0, -Y]$	$\left[0, 0, \frac{1}{2}\right]$	`[21,-23,-45,47]`
12	$[X, 0, Y]$	$\left[0, 0, \frac{1}{2}\right]$	`[22,-24,-46,48]`

Table 10: Wyckoff bond: `12f@6b`, bond symmetry: `2/m..`

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-4,-25,28]`

continued ...

Table 10

No.	vector	center	mapping
2	$[0, -X, Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-3,-26,27]`
3	$[Y, 0, X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-8,-29,32]`
4	$[Y, 0, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[6,-7,-30,31]`
5	$[X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-12,-33,36]`
6	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[10,-11,-34,35]`
7	$[X, 0, -Y]$	$\left[0, \frac{1}{2}, 0\right]$	`[13,-16,-37,40]`
8	$[-X, 0, -Y]$	$\left[0, \frac{1}{2}, 0\right]$	`[14,-15,-38,39]`
9	$[0, Y, -X]$	$\left[\frac{1}{2}, 0, 0\right]$	`[17,-20,-41,44]`
10	$[0, Y, X]$	$\left[\frac{1}{2}, 0, 0\right]$	`[18,-19,-42,43]`
11	$[Y, X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[21,-24,-45,48]`
12	$[Y, -X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[22,-23,-46,47]`

Table 11: Wyckoff bond: `24g@6b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-25]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-26]`
3	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-27]`
4	$[X, -Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,-28]`
5	$[Z, X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-29]`
6	$[Z, -X, -Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[6,-30]`
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[7,-31]`
8	$[-Z, X, -Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[8,-32]`
9	$[Y, Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-33]`
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[10,-34]`
11	$[Y, -Z, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[11,-35]`
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[12,-36]`
13	$[Y, X, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[13,-37]`
14	$[-Y, -X, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[14,-38]`
15	$[Y, -X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[15,-39]`
16	$[-Y, X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[16,-40]`
17	$[X, Z, -Y]$	$\left[\frac{1}{2}, 0, 0\right]$	`[17,-41]`
18	$[-X, Z, Y]$	$\left[\frac{1}{2}, 0, 0\right]$	`[18,-42]`
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}, 0, 0\right]$	`[19,-43]`
20	$[X, -Z, Y]$	$\left[\frac{1}{2}, 0, 0\right]$	`[20,-44]`
21	$[Z, Y, -X]$	$\left[0, 0, \frac{1}{2}\right]$	`[21,-45]`
22	$[Z, -Y, X]$	$\left[0, 0, \frac{1}{2}\right]$	`[22,-46]`
23	$[-Z, Y, X]$	$\left[0, 0, \frac{1}{2}\right]$	`[23,-47]`
24	$[-Z, -Y, -X]$	$\left[0, 0, \frac{1}{2}\right]$	`[24,-48]`

* Wyckoff site: `6c`, site symmetry: `-4m.2`

Table 12: Wyckoff bond: `6a@6c`, bond symmetry: `-4m.2`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[1,4,-18,-19,26,27,-41,-44]`
2	$[-X, 0, 0]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[2,3,-17,-20,25,28,-42,-43]`
3	$[0, X, 0]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[5,8,-14,-15,30,31,-37,-40]`
4	$[0, -X, 0]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[6,7,-13,-16,29,32,-38,-39]`
5	$[0, 0, X]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[9,12,-21,-24,34,35,-46,-47]`
6	$[0, 0, -X]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[10,11,-22,-23,33,36,-45,-48]`

Table 13: Wyckoff bond: `12b@6c`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[0, X, X]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[1,-4,18,-19]`
2	$[0, -X, X]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[2,-3,-17,20]`
3	$[X, 0, X]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[5,-8,-14,15]`
4	$[X, 0, -X]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[6,-7,13,-16]`
5	$[X, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[9,-12,21,-24]`
6	$[-X, X, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[10,-11,-22,23]`
7	$[0, -X, -X]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[25,-28,42,-43]`
8	$[0, X, -X]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[26,-27,-41,44]`
9	$[-X, 0, -X]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[29,-32,-38,39]`
10	$[-X, 0, X]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[30,-31,37,-40]`
11	$[-X, -X, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[33,-36,45,-48]`
12	$[X, -X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[34,-35,-46,47]`

Table 14: Wyckoff bond: `12c@6c`, bond symmetry: `mm2..`

No.	vector	center	mapping
1	$[0, 0, X]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[1,-4,-26,27]`
2	$[0, 0, X]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[2,-3,-25,28]`
3	$[X, 0, 0]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[5,-8,-30,31]`
4	$[X, 0, 0]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[6,-7,-29,32]`
5	$[0, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[9,-12,-34,35]`
6	$[0, X, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[10,-11,-33,36]`
7	$[0, 0, -X]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[13,-16,-38,39]`
8	$[0, 0, -X]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[14,-15,-37,40]`
9	$[0, X, 0]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[17,-20,-42,43]`
10	$[0, X, 0]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[18,-19,-41,44]`
11	$[X, 0, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[21,-24,-46,47]`
12	$[X, 0, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[22,-23,-45,48]`

Table 15: Wyckoff bond: `24d@6c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[Y, X, X]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[1,-19]`

continued ...

Table 15

No.	vector	center	mapping
2	$[-Y, -X, X]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[2,-17]`
3	$[-Y, X, -X]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[3,-20]`
4	$[Y, -X, -X]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[4,-18]`
5	$[X, Y, X]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[5,-14]`
6	$[X, -Y, -X]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[6,-16]`
7	$[-X, -Y, X]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[7,-13]`
8	$[-X, Y, -X]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[8,-15]`
9	$[X, X, Y]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[9,-24]`
10	$[-X, X, -Y]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[10,-22]`
11	$[X, -X, -Y]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[11,-23]`
12	$[-X, -X, Y]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[12,-21]`
13	$[-Y, -X, -X]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[25,-43]`
14	$[Y, X, -X]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[26,-41]`
15	$[Y, -X, X]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[27,-44]`
16	$[-Y, X, X]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[28,-42]`
17	$[-X, -Y, -X]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[29,-38]`
18	$[-X, Y, X]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[30,-40]`
19	$[X, Y, -X]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[31,-37]`
20	$[X, -Y, X]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[32,-39]`
21	$[-X, -X, -Y]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[33,-48]`
22	$[X, -X, Y]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[34,-46]`
23	$[-X, X, Y]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[35,-47]`
24	$[X, X, -Y]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[36,-45]`

Table 16: Wyckoff bond: `24e@6c`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[1,26]`
2	$[-X, -Y, 0]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[2,25]`
3	$[-X, Y, 0]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[3,28]`
4	$[X, -Y, 0]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[4,27]`
5	$[0, X, Y]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[5,30]`
6	$[0, -X, -Y]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[6,29]`
7	$[0, -X, Y]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[7,32]`
8	$[0, X, -Y]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[8,31]`
9	$[Y, 0, X]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[9,34]`
10	$[-Y, 0, -X]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[10,33]`
11	$[Y, 0, -X]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[11,36]`
12	$[-Y, 0, X]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[12,35]`
13	$[Y, X, 0]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[13,38]`
14	$[-Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[14,37]`
15	$[Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[15,40]`
16	$[-Y, X, 0]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[16,39]`
17	$[X, 0, -Y]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[17,42]`
18	$[-X, 0, Y]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[18,41]`

continued ...

Table 16

No.	vector	center	mapping
19	$[-X, 0, -Y]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[19,44]
20	$[X, 0, Y]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[20,43]
21	$[0, Y, -X]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[21,46]
22	$[0, -Y, X]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[22,45]
23	$[0, Y, X]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[23,48]
24	$[0, -Y, -X]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[24,47]

Table 17: Wyckoff bond: 24f@6c, bond symmetry: 2..

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[1,-4]
2	$[0, -X, Y]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[2,-3]
3	$[Y, 0, X]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[5,-8]
4	$[Y, 0, -X]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[6,-7]
5	$[X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[9,-12]
6	$[-X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[10,-11]
7	$[X, 0, -Y]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[13,-16]
8	$[-X, 0, -Y]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[14,-15]
9	$[0, Y, -X]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[17,-20]
10	$[0, Y, X]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[18,-19]
11	$[Y, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[21,-24]
12	$[Y, -X, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[22,-23]
13	$[0, -X, -Y]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[25,-28]
14	$[0, X, -Y]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[26,-27]
15	$[-Y, 0, -X]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[29,-32]
16	$[-Y, 0, X]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[30,-31]
17	$[-X, -Y, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[33,-36]
18	$[X, -Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[34,-35]
19	$[-X, 0, Y]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[37,-40]
20	$[X, 0, Y]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[38,-39]
21	$[0, -Y, X]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[41,-44]
22	$[0, -Y, -X]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[42,-43]
23	$[-Y, -X, 0]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[45,-48]
24	$[-Y, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[46,-47]

Table 18: Wyckoff bond: 48g@6c, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[3]
4	$[X, -Y, -Z]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[4]
5	$[Z, X, Y]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[5]
6	$[Z, -X, -Y]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[6]

continued ...

Table 18

No.	vector	center	mapping
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[8]
9	$[Y, Z, X]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[9]
10	$[-Y, Z, -X]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[10]
11	$[Y, -Z, -X]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[11]
12	$[-Y, -Z, X]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[12]
13	$[Y, X, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[14]
15	$[Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[16]
17	$[X, Z, -Y]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[19]
20	$[X, -Z, Y]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[20]
21	$[Z, Y, -X]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[21]
22	$[Z, -Y, X]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[22]
23	$[-Z, Y, X]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[25]
26	$[X, Y, -Z]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[26]
27	$[X, -Y, Z]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[27]
28	$[-X, Y, Z]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[30]
31	$[Z, X, -Y]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[31]
32	$[Z, -X, Y]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[33]
34	$[Y, -Z, X]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[34]
35	$[-Y, Z, X]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[35]
36	$[Y, Z, -X]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[37]
38	$[Y, X, Z]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	[39]
40	$[Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[42]
43	$[X, Z, Y]$	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	[44]
45	$[-Z, -Y, X]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[46]
47	$[Z, -Y, -X]$	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[47]
48	$[Z, Y, X]$	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[48]

* Wyckoff site: 6d, site symmetry: -4m.2

Table 19: Wyckoff bond: `6a@6d`, bond symmetry: `-4m.2`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[1,4,-18,-19,26,27,-41,-44]`
2	$[-X, 0, 0]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[2,3,-17,-20,25,28,-42,-43]`
3	$[0, X, 0]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[5,8,-14,-15,30,31,-37,-40]`
4	$[0, -X, 0]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[6,7,-13,-16,29,32,-38,-39]`
5	$[0, 0, X]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[9,12,-21,-24,34,35,-46,-47]`
6	$[0, 0, -X]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[10,11,-22,-23,33,36,-45,-48]`

Table 20: Wyckoff bond: `12b@6d`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[0, X, X]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[1,-4,18,-19]`
2	$[0, -X, X]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[2,-3,-17,20]`
3	$[X, 0, X]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[5,-8,-14,15]`
4	$[X, 0, -X]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[6,-7,13,-16]`
5	$[X, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[9,-12,21,-24]`
6	$[-X, X, 0]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[10,-11,-22,23]`
7	$[0, -X, -X]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[25,-28,42,-43]`
8	$[0, X, -X]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[26,-27,-41,44]`
9	$[-X, 0, -X]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[29,-32,-38,39]`
10	$[-X, 0, X]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[30,-31,37,-40]`
11	$[-X, -X, 0]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[33,-36,45,-48]`
12	$[X, -X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[34,-35,-46,47]`

Table 21: Wyckoff bond: `12c@6d`, bond symmetry: `mm2..`

No.	vector	center	mapping
1	$[0, 0, X]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[1,-4,-26,27]`
2	$[0, 0, X]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[2,-3,-25,28]`
3	$[X, 0, 0]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[5,-8,-30,31]`
4	$[X, 0, 0]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[6,-7,-29,32]`
5	$[0, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[9,-12,-34,35]`
6	$[0, X, 0]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[10,-11,-33,36]`
7	$[0, 0, -X]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[13,-16,-38,39]`
8	$[0, 0, -X]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[14,-15,-37,40]`
9	$[0, X, 0]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[17,-20,-42,43]`
10	$[0, X, 0]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[18,-19,-41,44]`
11	$[X, 0, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[21,-24,-46,47]`
12	$[X, 0, 0]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[22,-23,-45,48]`

Table 22: Wyckoff bond: `24d@6d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[Y, X, X]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[1,-19]`

continued ...

Table 22

No.	vector	center	mapping
2	$[-Y, -X, X]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[2,-17]
3	$[-Y, X, -X]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[3,-20]
4	$[Y, -X, -X]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[4,-18]
5	$[X, Y, X]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[5,-14]
6	$[X, -Y, -X]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[6,-16]
7	$[-X, -Y, X]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[7,-13]
8	$[-X, Y, -X]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[8,-15]
9	$[X, X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[9,-24]
10	$[-X, X, -Y]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[10,-22]
11	$[X, -X, -Y]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[11,-23]
12	$[-X, -X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[12,-21]
13	$[-Y, -X, -X]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[25,-43]
14	$[Y, X, -X]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[26,-41]
15	$[Y, -X, X]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[27,-44]
16	$[-Y, X, X]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[28,-42]
17	$[-X, -Y, -X]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[29,-38]
18	$[-X, Y, X]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[30,-40]
19	$[X, Y, -X]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[31,-37]
20	$[X, -Y, X]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[32,-39]
21	$[-X, -X, -Y]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[33,-48]
22	$[X, -X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[34,-46]
23	$[-X, X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[35,-47]
24	$[X, X, -Y]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[36,-45]

Table 23: Wyckoff bond: 24e@6d, bond symmetry: m..

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[1,26]
2	$[-X, -Y, 0]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[2,25]
3	$[-X, Y, 0]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[3,28]
4	$[X, -Y, 0]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[4,27]
5	$[0, X, Y]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[5,30]
6	$[0, -X, -Y]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[6,29]
7	$[0, -X, Y]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[7,32]
8	$[0, X, -Y]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[8,31]
9	$[Y, 0, X]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[9,34]
10	$[-Y, 0, -X]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[10,33]
11	$[Y, 0, -X]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[11,36]
12	$[-Y, 0, X]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[12,35]
13	$[Y, X, 0]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[13,38]
14	$[-Y, -X, 0]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[14,37]
15	$[Y, -X, 0]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[15,40]
16	$[-Y, X, 0]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[16,39]
17	$[X, 0, -Y]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[17,42]
18	$[-X, 0, Y]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[18,41]

continued ...

Table 23

No.	vector	center	mapping
19	$[-X, 0, -Y]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[19,44]
20	$[X, 0, Y]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[20,43]
21	$[0, Y, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[21,46]
22	$[0, -Y, X]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[22,45]
23	$[0, Y, X]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[23,48]
24	$[0, -Y, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[24,47]

Table 24: Wyckoff bond: 24f@6d, bond symmetry: 2..

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[1,-4]
2	$[0, -X, Y]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[2,-3]
3	$[Y, 0, X]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[5,-8]
4	$[Y, 0, -X]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[6,-7]
5	$[X, Y, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[9,-12]
6	$[-X, Y, 0]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[10,-11]
7	$[X, 0, -Y]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[13,-16]
8	$[-X, 0, -Y]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[14,-15]
9	$[0, Y, -X]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[17,-20]
10	$[0, Y, X]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[18,-19]
11	$[Y, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[21,-24]
12	$[Y, -X, 0]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[22,-23]
13	$[0, -X, -Y]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[25,-28]
14	$[0, X, -Y]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[26,-27]
15	$[-Y, 0, -X]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[29,-32]
16	$[-Y, 0, X]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[30,-31]
17	$[-X, -Y, 0]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[33,-36]
18	$[X, -Y, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[34,-35]
19	$[-X, 0, Y]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[37,-40]
20	$[X, 0, Y]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[38,-39]
21	$[0, -Y, X]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[41,-44]
22	$[0, -Y, -X]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[42,-43]
23	$[-Y, -X, 0]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[45,-48]
24	$[-Y, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[46,-47]

Table 25: Wyckoff bond: 48g@6d, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[3]
4	$[X, -Y, -Z]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[4]
5	$[Z, X, Y]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[5]
6	$[Z, -X, -Y]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[6]

continued ...

Table 25

No.	vector	center	mapping
7	$[-Z, -X, Y]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[8]
9	$[Y, Z, X]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[10]
11	$[Y, -Z, -X]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[12]
13	$[Y, X, -Z]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[14]
15	$[Y, -X, Z]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[15]
16	$[-Y, X, Z]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[16]
17	$[X, Z, -Y]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[19]
20	$[X, -Z, Y]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[20]
21	$[Z, Y, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[25]
26	$[X, Y, -Z]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[26]
27	$[X, -Y, Z]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[27]
28	$[-X, Y, Z]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[29]
30	$[-Z, X, Y]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[30]
31	$[Z, X, -Y]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[31]
32	$[Z, -X, Y]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[33]
34	$[Y, -Z, X]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[34]
35	$[-Y, Z, X]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[35]
36	$[Y, Z, -X]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[36]
37	$[-Y, -X, Z]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[37]
38	$[Y, X, Z]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[38]
39	$[-Y, X, -Z]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[41]
42	$[X, -Z, -Y]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[42]
43	$[X, Z, Y]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[46]
47	$[Z, -Y, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[47]
48	$[Z, Y, X]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[48]

* Wyckoff site: 8e, site symmetry: .32

Table 26: Wyckoff bond: `8a@8e`, bond symmetry: `.32`

No.	vector	center	mapping
1	$[X, X, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,5,9,-14,-19,-24]`
2	$[-X, -X, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[2,7,12,-13,-17,-21]`
3	$[-X, X, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[3,8,10,-15,-20,-22]`
4	$[X, -X, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[4,6,11,-16,-18,-23]`
5	$[-X, -X, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[25,29,33,-38,-43,-48]`
6	$[X, X, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[26,31,36,-37,-41,-45]`
7	$[X, -X, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[27,32,34,-39,-44,-46]`
8	$[-X, X, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[28,30,35,-40,-42,-47]`

Table 27: Wyckoff bond: `24b@8e`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,-14]`
2	$[-X, -X, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[2,-13]`
3	$[-X, X, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[3,-15]`
4	$[X, -X, -Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[4,-16]`
5	$[Y, X, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[5,-24]`
6	$[Y, -X, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[6,-23]`
7	$[-Y, -X, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[7,-21]`
8	$[-Y, X, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[8,-22]`
9	$[X, Y, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[9,-19]`
10	$[-X, Y, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[10,-20]`
11	$[X, -Y, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[11,-18]`
12	$[-X, -Y, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[12,-17]`
13	$[-X, -X, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[25,-38]`
14	$[X, X, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[26,-37]`
15	$[X, -X, Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[27,-39]`
16	$[-X, X, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[28,-40]`
17	$[-Y, -X, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[29,-48]`
18	$[-Y, X, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[30,-47]`
19	$[Y, X, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[31,-45]`
20	$[Y, -X, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[32,-46]`
21	$[-X, -Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[33,-43]`
22	$[X, -Y, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[34,-44]`
23	$[-X, Y, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[35,-42]`
24	$[X, Y, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[36,-41]`

Table 28: Wyckoff bond: `24c@8e`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,14]`
2	$[-X, X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[2,13]`
3	$[-X, -X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[3,15]`
4	$[X, X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[4,16]`

continued ...

Table 28

No.	vector	center	mapping
5	$[0, X, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[5,24]
6	$[0, -X, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[6,23]
7	$[0, -X, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[7,21]
8	$[0, X, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	[8,22]
9	$[-X, 0, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[9,19]
10	$[X, 0, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	[10,20]
11	$[-X, 0, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[11,18]
12	$[X, 0, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[12,17]
13	$[-X, X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[25,38]
14	$[X, -X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	[26,37]
15	$[X, X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	[27,39]
16	$[-X, -X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[28,40]
17	$[0, -X, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[29,48]
18	$[0, X, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[30,47]
19	$[0, X, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	[31,45]
20	$[0, -X, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	[32,46]
21	$[X, 0, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[33,43]
22	$[-X, 0, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	[34,44]
23	$[X, 0, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[35,42]
24	$[-X, 0, -X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	[36,41]

Table 29: Wyckoff bond: 48d@8e, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[1]
2	$[-X, -Y, Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[2]
3	$[-X, Y, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	[3]
4	$[X, -Y, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[4]
5	$[Z, X, Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[5]
6	$[Z, -X, -Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[6]
7	$[-Z, -X, Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[7]
8	$[-Z, X, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	[8]
9	$[Y, Z, X]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[9]
10	$[-Y, Z, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	[10]
11	$[Y, -Z, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[11]
12	$[-Y, -Z, X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[12]
13	$[Y, X, -Z]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[13]
14	$[-Y, -X, -Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[14]
15	$[Y, -X, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	[15]
16	$[-Y, X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[16]
17	$[X, Z, -Y]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[17]
18	$[-X, Z, Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[18]
19	$[-X, -Z, -Y]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[19]
20	$[X, -Z, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	[20]
21	$[Z, Y, -X]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[21]

continued ...

Table 29

No.	vector	center	mapping
22	$[Z, -Y, X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[25]
26	$[X, Y, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[26]
27	$[X, -Y, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[27]
28	$[-X, Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[30]
31	$[Z, X, -Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[31]
32	$[Z, -X, Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[33]
34	$[Y, -Z, X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[34]
35	$[-Y, Z, X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[35]
36	$[Y, Z, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[37]
38	$[Y, X, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[42]
43	$[X, Z, Y]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[46]
47	$[Z, -Y, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[47]
48	$[Z, Y, X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[48]

* Wyckoff site: 12f, site symmetry: mm2..

Table 30: Wyckoff bond: 12a@12f, bond symmetry: mm2..

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 0, 0]$	[1,4,26,27]
2	$[-X, 0, 0]$	$[-x, 0, 0]$	[2,3,25,28]
3	$[0, X, 0]$	$[0, x, 0]$	[5,8,30,31]
4	$[0, -X, 0]$	$[0, -x, 0]$	[6,7,29,32]
5	$[0, 0, X]$	$[0, 0, x]$	[9,12,34,35]
6	$[0, 0, -X]$	$[0, 0, -x]$	[10,11,33,36]
7	$[0, X, 0]$	$\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	[13,16,38,39]
8	$[0, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	[14,15,37,40]
9	$[X, 0, 0]$	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[17,20,42,43]
10	$[-X, 0, 0]$	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$	[18,19,41,44]
11	$[0, 0, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-x\right]$	[21,24,46,47]
12	$[0, 0, X]$	$\left[\frac{1}{2}, \frac{1}{2}, x+\frac{1}{2}\right]$	[22,23,45,48]

Table 31: Wyckoff bond: `12b@12f`, bond symmetry: `mm2..`

No.	vector	center	mapping
1	$[0,\ 0,\ X]$	$[x,\ 0,\ 0]$	`[1,-4,-26,27]`
2	$[0,\ 0,\ X]$	$[-x,\ 0,\ 0]$	`[2,-3,-25,28]`
3	$[X,\ 0,\ 0]$	$[0,\ x,\ 0]$	`[5,-8,-30,31]`
4	$[X,\ 0,\ 0]$	$[0,\ -x,\ 0]$	`[6,-7,-29,32]`
5	$[0,\ X,\ 0]$	$[0,\ 0,\ x]$	`[9,-12,-34,35]`
6	$[0,\ X,\ 0]$	$[0,\ 0,\ -x]$	`[10,-11,-33,36]`
7	$[0,\ 0,\ -X]$	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[13,-16,-38,39]`
8	$[0,\ 0,\ -X]$	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[14,-15,-37,40]`
9	$[0,\ X,\ 0]$	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[17,-20,-42,43]`
10	$[0,\ X,\ 0]$	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[18,-19,-41,44]`
11	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x\right]$	`[21,-24,-46,47]`
12	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}\right]$	`[22,-23,-45,48]`

Table 32: Wyckoff bond: `12c@12f`, bond symmetry: `mm2..`

No.	vector	center	mapping
1	$[0,\ X,\ 0]$	$[x,\ 0,\ 0]$	`[1,-4,26,-27]`
2	$[0,\ -X,\ 0]$	$[-x,\ 0,\ 0]$	`[2,-3,25,-28]`
3	$[0,\ 0,\ X]$	$[0,\ x,\ 0]$	`[5,-8,30,-31]`
4	$[0,\ 0,\ -X]$	$[0,\ -x,\ 0]$	`[6,-7,29,-32]`
5	$[X,\ 0,\ 0]$	$[0,\ 0,\ x]$	`[9,-12,34,-35]`
6	$[-X,\ 0,\ 0]$	$[0,\ 0,\ -x]$	`[10,-11,33,-36]`
7	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[13,-16,38,-39]`
8	$[-X,\ 0,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[14,-15,37,-40]`
9	$[0,\ 0,\ -X]$	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[17,-20,42,-43]`
10	$[0,\ 0,\ X]$	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[18,-19,41,-44]`
11	$[0,\ X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x\right]$	`[21,-24,46,-47]`
12	$[0,\ -X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}\right]$	`[22,-23,45,-48]`

Table 33: Wyckoff bond: `24d@12f`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$[x,\ 0,\ 0]$	`[1,26]`
2	$[-X,\ -Y,\ 0]$	$[-x,\ 0,\ 0]$	`[2,25]`
3	$[-X,\ Y,\ 0]$	$[-x,\ 0,\ 0]$	`[3,28]`
4	$[X,\ -Y,\ 0]$	$[x,\ 0,\ 0]$	`[4,27]`
5	$[0,\ X,\ Y]$	$[0,\ x,\ 0]$	`[5,30]`
6	$[0,\ -X,\ -Y]$	$[0,\ -x,\ 0]$	`[6,29]`
7	$[0,\ -X,\ Y]$	$[0,\ -x,\ 0]$	`[7,32]`
8	$[0,\ X,\ -Y]$	$[0,\ x,\ 0]$	`[8,31]`
9	$[Y,\ 0,\ X]$	$[0,\ 0,\ x]$	`[9,34]`
10	$[-Y,\ 0,\ -X]$	$[0,\ 0,\ -x]$	`[10,33]`
11	$[Y,\ 0,\ -X]$	$[0,\ 0,\ -x]$	`[11,36]`
12	$[-Y,\ 0,\ X]$	$[0,\ 0,\ x]$	`[12,35]`

continued ...

Table 33

No.	vector	center	mapping
13	$[Y,\ X,\ 0]$	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[13,38]`
14	$[-Y,\ -X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[14,37]`
15	$[Y,\ -X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[15,40]`
16	$[-Y,\ X,\ 0]$	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[16,39]`
17	$[X,\ 0,\ -Y]$	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[17,42]`
18	$[-X,\ 0,\ Y]$	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[18,41]`
19	$[-X,\ 0,\ -Y]$	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[19,44]`
20	$[X,\ 0,\ Y]$	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[20,43]`
21	$[0,\ Y,\ -X]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x\right]$	`[21,46]`
22	$[0,\ -Y,\ X]$	$\left[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}\right]$	`[22,45]`
23	$[0,\ Y,\ X]$	$\left[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}\right]$	`[23,48]`
24	$[0,\ -Y,\ -X]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x\right]$	`[24,47]`

Table 34: Wyckoff bond: `24e@12f`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[Y,\ 0,\ X]$	$[x,\ 0,\ 0]$	`[1,27]`
2	$[-Y,\ 0,\ X]$	$[-x,\ 0,\ 0]$	`[2,28]`
3	$[-Y,\ 0,\ -X]$	$[-x,\ 0,\ 0]$	`[3,25]`
4	$[Y,\ 0,\ -X]$	$[x,\ 0,\ 0]$	`[4,26]`
5	$[X,\ Y,\ 0]$	$[0,\ x,\ 0]$	`[5,31]`
6	$[X,\ -Y,\ 0]$	$[0,\ -x,\ 0]$	`[6,32]`
7	$[-X,\ -Y,\ 0]$	$[0,\ -x,\ 0]$	`[7,29]`
8	$[-X,\ Y,\ 0]$	$[0,\ x,\ 0]$	`[8,30]`
9	$[0,\ X,\ Y]$	$[0,\ 0,\ x]$	`[9,35]`
10	$[0,\ X,\ -Y]$	$[0,\ 0,\ -x]$	`[10,36]`
11	$[0,\ -X,\ -Y]$	$[0,\ 0,\ -x]$	`[11,33]`
12	$[0,\ -X,\ Y]$	$[0,\ 0,\ x]$	`[12,34]`
13	$[0,\ Y,\ -X]$	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[13,39]`
14	$[0,\ -Y,\ -X]$	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[14,40]`
15	$[0,\ -Y,\ X]$	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[15,37]`
16	$[0,\ Y,\ X]$	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[16,38]`
17	$[Y,\ X,\ 0]$	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[17,43]`
18	$[-Y,\ X,\ 0]$	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[18,44]`
19	$[-Y,\ -X,\ 0]$	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[19,41]`
20	$[Y,\ -X,\ 0]$	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[20,42]`
21	$[X,\ 0,\ -Y]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x\right]$	`[21,47]`
22	$[X,\ 0,\ Y]$	$\left[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}\right]$	`[22,48]`
23	$[-X,\ 0,\ Y]$	$\left[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}\right]$	`[23,45]`
24	$[-X,\ 0,\ -Y]$	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x\right]$	`[24,46]`

Table 35: Wyckoff bond: `24f@12f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\ X,\ Y]$	$[x,\ 0,\ 0]$	`[1,-4]`
2	$[0,\ -X,\ Y]$	$[-x,\ 0,\ 0]$	`[2,-3]`
3	$[Y,\ 0,\ X]$	$[0,\ x,\ 0]$	`[5,-8]`
4	$[Y,\ 0,\ -X]$	$[0,\ -x,\ 0]$	`[6,-7]`
5	$[X,\ Y,\ 0]$	$[0,\ 0,\ x]$	`[9,-12]`
6	$[-X,\ Y,\ 0]$	$[0,\ 0,\ -x]$	`[10,-11]`
7	$[X,\ 0,\ -Y]$	$[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}]$	`[13,-16]`
8	$[-X,\ 0,\ -Y]$	$[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}]$	`[14,-15]`
9	$[0,\ Y,\ -X]$	$[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}]$	`[17,-20]`
10	$[0,\ Y,\ X]$	$[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}]$	`[18,-19]`
11	$[Y,\ X,\ 0]$	$[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x]$	`[21,-24]`
12	$[Y,\ -X,\ 0]$	$[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}]$	`[22,-23]`
13	$[0,\ -X,\ -Y]$	$[-x,\ 0,\ 0]$	`[25,-28]`
14	$[0,\ X,\ -Y]$	$[x,\ 0,\ 0]$	`[26,-27]`
15	$[-Y,\ 0,\ -X]$	$[0,\ -x,\ 0]$	`[29,-32]`
16	$[-Y,\ 0,\ X]$	$[0,\ x,\ 0]$	`[30,-31]`
17	$[-X,\ -Y,\ 0]$	$[0,\ 0,\ -x]$	`[33,-36]`
18	$[X,\ -Y,\ 0]$	$[0,\ 0,\ x]$	`[34,-35]`
19	$[-X,\ 0,\ Y]$	$[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}]$	`[37,-40]`
20	$[X,\ 0,\ Y]$	$[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}]$	`[38,-39]`
21	$[0,\ -Y,\ X]$	$[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}]$	`[41,-44]`
22	$[0,\ -Y,\ -X]$	$[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}]$	`[42,-43]`
23	$[-Y,\ -X,\ 0]$	$[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}]$	`[45,-48]`
24	$[-Y,\ X,\ 0]$	$[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x]$	`[46,-47]`

Table 36: Wyckoff bond: `48g@12f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ 0,\ 0]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$[-x,\ 0,\ 0]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$[-x,\ 0,\ 0]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$[x,\ 0,\ 0]$	`[4]`
5	$[Z,\ X,\ Y]$	$[0,\ x,\ 0]$	`[5]`
6	$[Z,\ -X,\ -Y]$	$[0,\ -x,\ 0]$	`[6]`
7	$[-Z,\ -X,\ Y]$	$[0,\ -x,\ 0]$	`[7]`
8	$[-Z,\ X,\ -Y]$	$[0,\ x,\ 0]$	`[8]`
9	$[Y,\ Z,\ X]$	$[0,\ 0,\ x]$	`[9]`
10	$[-Y,\ Z,\ -X]$	$[0,\ 0,\ -x]$	`[10]`
11	$[Y,\ -Z,\ -X]$	$[0,\ 0,\ -x]$	`[11]`
12	$[-Y,\ -Z,\ X]$	$[0,\ 0,\ x]$	`[12]`
13	$[Y,\ X,\ -Z]$	$[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}]$	`[13]`
14	$[-Y,\ -X,\ -Z]$	$[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}]$	`[14]`
15	$[Y,\ -X,\ Z]$	$[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}]$	`[15]`
16	$[-Y,\ X,\ Z]$	$[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}]$	`[16]`
17	$[X,\ Z,\ -Y]$	$[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}]$	`[17]`

continued ...

Table 36

No.	vector	center	mapping
18	$[-X, Z, Y]$	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$	[19]
20	$[X, -Z, Y]$	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[20]
21	$[Z, Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-x\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, x+\frac{1}{2}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, x+\frac{1}{2}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-x\right]$	[24]
25	$[-X, -Y, -Z]$	$[-x, 0, 0]$	[25]
26	$[X, Y, -Z]$	$[x, 0, 0]$	[26]
27	$[X, -Y, Z]$	$[x, 0, 0]$	[27]
28	$[-X, Y, Z]$	$[-x, 0, 0]$	[28]
29	$[-Z, -X, -Y]$	$[0, -x, 0]$	[29]
30	$[-Z, X, Y]$	$[0, x, 0]$	[30]
31	$[Z, X, -Y]$	$[0, x, 0]$	[31]
32	$[Z, -X, Y]$	$[0, -x, 0]$	[32]
33	$[-Y, -Z, -X]$	$[0, 0, -x]$	[33]
34	$[Y, -Z, X]$	$[0, 0, x]$	[34]
35	$[-Y, Z, X]$	$[0, 0, x]$	[35]
36	$[Y, Z, -X]$	$[0, 0, -x]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	[37]
38	$[Y, X, Z]$	$\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[42]
43	$[X, Z, Y]$	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, x+\frac{1}{2}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-x\right]$	[46]
47	$[Z, -Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-x\right]$	[47]
48	$[Z, Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, x+\frac{1}{2}\right]$	[48]

* Wyckoff site: 12g, site symmetry: mm2..

Table 37: Wyckoff bond: 12a@12g, bond symmetry: mm2..

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, 0, \frac{1}{2}\right]$	[1,4,26,27]
2	$[-X, 0, 0]$	$\left[-x, 0, \frac{1}{2}\right]$	[2,3,25,28]
3	$[0, X, 0]$	$\left[\frac{1}{2}, x, 0\right]$	[5,8,30,31]
4	$[0, -X, 0]$	$\left[\frac{1}{2}, -x, 0\right]$	[6,7,29,32]
5	$[0, 0, X]$	$\left[0, \frac{1}{2}, x\right]$	[9,12,34,35]
6	$[0, 0, -X]$	$\left[0, \frac{1}{2}, -x\right]$	[10,11,33,36]
7	$[0, X, 0]$	$\left[\frac{1}{2}, x+\frac{1}{2}, 0\right]$	[13,16,38,39]
8	$[0, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}-x, 0\right]$	[14,15,37,40]
9	$[X, 0, 0]$	$\left[x+\frac{1}{2}, 0, \frac{1}{2}\right]$	[17,20,42,43]
10	$[-X, 0, 0]$	$\left[\frac{1}{2}-x, 0, \frac{1}{2}\right]$	[18,19,41,44]

continued ...

Table 37

No.	vector	center	mapping
11	$[0,\ 0,\ -X]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-x\right]$	`[21,24,46,47]`
12	$[0,\ 0,\ X]$	$\left[0,\ \frac{1}{2},\ x+\frac{1}{2}\right]$	`[22,23,45,48]`

Table 38: Wyckoff bond: `12b@12g`, bond symmetry: `mm2..`

No.	vector	center	mapping
1	$[0,\ 0,\ X]$	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,-4,-26,27]`
2	$[0,\ 0,\ X]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[2,-3,-25,28]`
3	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ x,\ 0\right]$	`[5,-8,-30,31]`
4	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[6,-7,-29,32]`
5	$[0,\ X,\ 0]$	$\left[0,\ \frac{1}{2},\ x\right]$	`[9,-12,-34,35]`
6	$[0,\ X,\ 0]$	$\left[0,\ \frac{1}{2},\ -x\right]$	`[10,-11,-33,36]`
7	$[0,\ 0,\ -X]$	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	`[13,-16,-38,39]`
8	$[0,\ 0,\ -X]$	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[14,-15,-37,40]`
9	$[0,\ X,\ 0]$	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[17,-20,-42,43]`
10	$[0,\ X,\ 0]$	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	`[18,-19,-41,44]`
11	$[X,\ 0,\ 0]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-x\right]$	`[21,-24,-46,47]`
12	$[X,\ 0,\ 0]$	$\left[0,\ \frac{1}{2},\ x+\frac{1}{2}\right]$	`[22,-23,-45,48]`

Table 39: Wyckoff bond: `24c@12g`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,26]`
2	$[-X,\ -Y,\ 0]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[2,25]`
3	$[-X,\ Y,\ 0]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[3,28]`
4	$[X,\ -Y,\ 0]$	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[4,27]`
5	$[0,\ X,\ Y]$	$\left[\frac{1}{2},\ x,\ 0\right]$	`[5,30]`
6	$[0,\ -X,\ -Y]$	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[6,29]`
7	$[0,\ -X,\ Y]$	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[7,32]`
8	$[0,\ X,\ -Y]$	$\left[\frac{1}{2},\ x,\ 0\right]$	`[8,31]`
9	$[Y,\ 0,\ X]$	$\left[0,\ \frac{1}{2},\ x\right]$	`[9,34]`
10	$[-Y,\ 0,\ -X]$	$\left[0,\ \frac{1}{2},\ -x\right]$	`[10,33]`
11	$[Y,\ 0,\ -X]$	$\left[0,\ \frac{1}{2},\ -x\right]$	`[11,36]`
12	$[-Y,\ 0,\ X]$	$\left[0,\ \frac{1}{2},\ x\right]$	`[12,35]`
13	$[Y,\ X,\ 0]$	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	`[13,38]`
14	$[-Y,\ -X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[14,37]`
15	$[Y,\ -X,\ 0]$	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[15,40]`
16	$[-Y,\ X,\ 0]$	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	`[16,39]`
17	$[X,\ 0,\ -Y]$	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[17,42]`
18	$[-X,\ 0,\ Y]$	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	`[18,41]`
19	$[-X,\ 0,\ -Y]$	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	`[19,44]`
20	$[X,\ 0,\ Y]$	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[20,43]`
21	$[0,\ Y,\ -X]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-x\right]$	`[21,46]`
22	$[0,\ -Y,\ X]$	$\left[0,\ \frac{1}{2},\ x+\frac{1}{2}\right]$	`[22,45]`

continued ...

Table 39

No.	vector	center	mapping
23	$[0,\ Y,\ X]$	$\left[0,\ \frac{1}{2},\ x+\frac{1}{2}\right]$	`[23,48]`
24	$[0,\ -Y,\ -X]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-x\right]$	`[24,47]`

Table 40: Wyckoff bond: `24d@12g`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\ X,\ Y]$	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,-4]`
2	$[0,\ -X,\ Y]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[2,-3]`
3	$[Y,\ 0,\ X]$	$\left[\frac{1}{2},\ x,\ 0\right]$	`[5,-8]`
4	$[Y,\ 0,\ -X]$	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[6,-7]`
5	$[X,\ Y,\ 0]$	$\left[0,\ \frac{1}{2},\ x\right]$	`[9,-12]`
6	$[-X,\ Y,\ 0]$	$\left[0,\ \frac{1}{2},\ -x\right]$	`[10,-11]`
7	$[X,\ 0,\ -Y]$	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	`[13,-16]`
8	$[-X,\ 0,\ -Y]$	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[14,-15]`
9	$[0,\ Y,\ -X]$	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[17,-20]`
10	$[0,\ Y,\ X]$	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	`[18,-19]`
11	$[Y,\ X,\ 0]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-x\right]$	`[21,-24]`
12	$[Y,\ -X,\ 0]$	$\left[0,\ \frac{1}{2},\ x+\frac{1}{2}\right]$	`[22,-23]`
13	$[0,\ -X,\ -Y]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[25,-28]`
14	$[0,\ X,\ -Y]$	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[26,-27]`
15	$[-Y,\ 0,\ -X]$	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[29,-32]`
16	$[-Y,\ 0,\ X]$	$\left[\frac{1}{2},\ x,\ 0\right]$	`[30,-31]`
17	$[-X,\ -Y,\ 0]$	$\left[0,\ \frac{1}{2},\ -x\right]$	`[33,-36]`
18	$[X,\ -Y,\ 0]$	$\left[0,\ \frac{1}{2},\ x\right]$	`[34,-35]`
19	$[-X,\ 0,\ Y]$	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[37,-40]`
20	$[X,\ 0,\ Y]$	$\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	`[38,-39]`
21	$[0,\ -Y,\ X]$	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	`[41,-44]`
22	$[0,\ -Y,\ -X]$	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[42,-43]`
23	$[-Y,\ -X,\ 0]$	$\left[0,\ \frac{1}{2},\ x+\frac{1}{2}\right]$	`[45,-48]`
24	$[-Y,\ X,\ 0]$	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-x\right]$	`[46,-47]`

Table 41: Wyckoff bond: `48e@12g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[4]`
5	$[Z,\ X,\ Y]$	$\left[\frac{1}{2},\ x,\ 0\right]$	`[5]`
6	$[Z,\ -X,\ -Y]$	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[6]`
7	$[-Z,\ -X,\ Y]$	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[7]`
8	$[-Z,\ X,\ -Y]$	$\left[\frac{1}{2},\ x,\ 0\right]$	`[8]`
9	$[Y,\ Z,\ X]$	$\left[0,\ \frac{1}{2},\ x\right]$	`[9]`
10	$[-Y,\ Z,\ -X]$	$\left[0,\ \frac{1}{2},\ -x\right]$	`[10]`

continued ...

Table 41

No.	vector	center	mapping
11	$[Y, -Z, -X]$	$\left[0, \frac{1}{2}, -x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[0, \frac{1}{2}, x\right]$	[12]
13	$[Y, X, -Z]$	$\left[\frac{1}{2}, x+\frac{1}{2}, 0\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}-x, 0\right]$	[14]
15	$[Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}-x, 0\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{1}{2}, x+\frac{1}{2}, 0\right]$	[16]
17	$[X, Z, -Y]$	$\left[x+\frac{1}{2}, 0, \frac{1}{2}\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{2}-x, 0, \frac{1}{2}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}-x, 0, \frac{1}{2}\right]$	[19]
20	$[X, -Z, Y]$	$\left[x+\frac{1}{2}, 0, \frac{1}{2}\right]$	[20]
21	$[Z, Y, -X]$	$\left[0, \frac{1}{2}, \frac{1}{2}-x\right]$	[21]
22	$[Z, -Y, X]$	$\left[0, \frac{1}{2}, x+\frac{1}{2}\right]$	[22]
23	$[-Z, Y, X]$	$\left[0, \frac{1}{2}, x+\frac{1}{2}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[0, \frac{1}{2}, \frac{1}{2}-x\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[-x, 0, \frac{1}{2}\right]$	[25]
26	$[X, Y, -Z]$	$\left[x, 0, \frac{1}{2}\right]$	[26]
27	$[X, -Y, Z]$	$\left[x, 0, \frac{1}{2}\right]$	[27]
28	$[-X, Y, Z]$	$\left[-x, 0, \frac{1}{2}\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[\frac{1}{2}, -x, 0\right]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{1}{2}, x, 0\right]$	[30]
31	$[Z, X, -Y]$	$\left[\frac{1}{2}, x, 0\right]$	[31]
32	$[Z, -X, Y]$	$\left[\frac{1}{2}, -x, 0\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[0, \frac{1}{2}, -x\right]$	[33]
34	$[Y, -Z, X]$	$\left[0, \frac{1}{2}, x\right]$	[34]
35	$[-Y, Z, X]$	$\left[0, \frac{1}{2}, x\right]$	[35]
36	$[Y, Z, -X]$	$\left[0, \frac{1}{2}, -x\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}-x, 0\right]$	[37]
38	$[Y, X, Z]$	$\left[\frac{1}{2}, x+\frac{1}{2}, 0\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{2}, x+\frac{1}{2}, 0\right]$	[39]
40	$[Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}-x, 0\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{2}-x, 0, \frac{1}{2}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[x+\frac{1}{2}, 0, \frac{1}{2}\right]$	[42]
43	$[X, Z, Y]$	$\left[x+\frac{1}{2}, 0, \frac{1}{2}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{2}-x, 0, \frac{1}{2}\right]$	[44]
45	$[-Z, -Y, X]$	$\left[0, \frac{1}{2}, x+\frac{1}{2}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[0, \frac{1}{2}, \frac{1}{2}-x\right]$	[46]
47	$[Z, -Y, -X]$	$\left[0, \frac{1}{2}, \frac{1}{2}-x\right]$	[47]
48	$[Z, Y, X]$	$\left[0, \frac{1}{2}, x+\frac{1}{2}\right]$	[48]

* Wyckoff site: 12h, site symmetry: mm2..

Table 42: Wyckoff bond: 12a@12h, bond symmetry: mm2..

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, \frac{1}{2}, 0\right]$	[1,4,26,27]
2	$[-X, 0, 0]$	$\left[-x, \frac{1}{2}, 0\right]$	[2,3,25,28]
3	$[0, X, 0]$	$\left[0, x, \frac{1}{2}\right]$	[5,8,30,31]

continued ...

Table 42

No.	vector	center	mapping
4	$[0,\ -X,\ 0]$	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[6,7,29,32]`
5	$[0,\ 0,\ X]$	$\left[\frac{1}{2},\ 0,\ x\right]$	`[9,12,34,35]`
6	$[0,\ 0,\ -X]$	$\left[\frac{1}{2},\ 0,\ -x\right]$	`[10,11,33,36]`
7	$[0,\ X,\ 0]$	$\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[13,16,38,39]`
8	$[0,\ -X,\ 0]$	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[14,15,37,40]`
9	$[X,\ 0,\ 0]$	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[17,20,42,43]`
10	$[-X,\ 0,\ 0]$	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	`[18,19,41,44]`
11	$[0,\ 0,\ -X]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-x\right]$	`[21,24,46,47]`
12	$[0,\ 0,\ X]$	$\left[\frac{1}{2},\ 0,\ x+\frac{1}{2}\right]$	`[22,23,45,48]`

Table 43: Wyckoff bond: `12b@12h`, bond symmetry: `mm2..`

No.	vector	center	mapping
1	$[0,\ 0,\ X]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,-4,-26,27]`
2	$[0,\ 0,\ X]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[2,-3,-25,28]`
3	$[X,\ 0,\ 0]$	$\left[0,\ x,\ \frac{1}{2}\right]$	`[5,-8,-30,31]`
4	$[X,\ 0,\ 0]$	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[6,-7,-29,32]`
5	$[0,\ X,\ 0]$	$\left[\frac{1}{2},\ 0,\ x\right]$	`[9,-12,-34,35]`
6	$[0,\ X,\ 0]$	$\left[\frac{1}{2},\ 0,\ -x\right]$	`[10,-11,-33,36]`
7	$[0,\ 0,\ -X]$	$\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[13,-16,-38,39]`
8	$[0,\ 0,\ -X]$	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[14,-15,-37,40]`
9	$[0,\ X,\ 0]$	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[17,-20,-42,43]`
10	$[0,\ X,\ 0]$	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	`[18,-19,-41,44]`
11	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-x\right]$	`[21,-24,-46,47]`
12	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ 0,\ x+\frac{1}{2}\right]$	`[22,-23,-45,48]`

Table 44: Wyckoff bond: `24c@12h`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,26]`
2	$[-X,\ -Y,\ 0]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[2,25]`
3	$[-X,\ Y,\ 0]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[3,28]`
4	$[X,\ -Y,\ 0]$	$\left[x,\ \frac{1}{2},\ 0\right]$	`[4,27]`
5	$[0,\ X,\ Y]$	$\left[0,\ x,\ \frac{1}{2}\right]$	`[5,30]`
6	$[0,\ -X,\ -Y]$	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[6,29]`
7	$[0,\ -X,\ Y]$	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[7,32]`
8	$[0,\ X,\ -Y]$	$\left[0,\ x,\ \frac{1}{2}\right]$	`[8,31]`
9	$[Y,\ 0,\ X]$	$\left[\frac{1}{2},\ 0,\ x\right]$	`[9,34]`
10	$[-Y,\ 0,\ -X]$	$\left[\frac{1}{2},\ 0,\ -x\right]$	`[10,33]`
11	$[Y,\ 0,\ -X]$	$\left[\frac{1}{2},\ 0,\ -x\right]$	`[11,36]`
12	$[-Y,\ 0,\ X]$	$\left[\frac{1}{2},\ 0,\ x\right]$	`[12,35]`
13	$[Y,\ X,\ 0]$	$\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[13,38]`
14	$[-Y,\ -X,\ 0]$	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[14,37]`
15	$[Y,\ -X,\ 0]$	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[15,40]`

continued ...

Table 44

No.	vector	center	mapping
16	$[-Y, X, 0]$	$\left[0, x+\frac{1}{2}, \frac{1}{2}\right]$	[16,39]
17	$[X, 0, -Y]$	$\left[x+\frac{1}{2}, \frac{1}{2}, 0\right]$	[17,42]
18	$[-X, 0, Y]$	$\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$	[18,41]
19	$[-X, 0, -Y]$	$\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$	[19,44]
20	$[X, 0, Y]$	$\left[x+\frac{1}{2}, \frac{1}{2}, 0\right]$	[20,43]
21	$[0, Y, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}-x\right]$	[21,46]
22	$[0, -Y, X]$	$\left[\frac{1}{2}, 0, x+\frac{1}{2}\right]$	[22,45]
23	$[0, Y, X]$	$\left[\frac{1}{2}, 0, x+\frac{1}{2}\right]$	[23,48]
24	$[0, -Y, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}-x\right]$	[24,47]

Table 45: Wyckoff bond: 24d@12h, bond symmetry: 2..

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[x, \frac{1}{2}, 0\right]$	[1,-4]
2	$[0, -X, Y]$	$\left[-x, \frac{1}{2}, 0\right]$	[2,-3]
3	$[Y, 0, X]$	$\left[0, x, \frac{1}{2}\right]$	[5,-8]
4	$[Y, 0, -X]$	$\left[0, -x, \frac{1}{2}\right]$	[6,-7]
5	$[X, Y, 0]$	$\left[\frac{1}{2}, 0, x\right]$	[9,-12]
6	$[-X, Y, 0]$	$\left[\frac{1}{2}, 0, -x\right]$	[10,-11]
7	$[X, 0, -Y]$	$\left[0, x+\frac{1}{2}, \frac{1}{2}\right]$	[13,-16]
8	$[-X, 0, -Y]$	$\left[0, \frac{1}{2}-x, \frac{1}{2}\right]$	[14,-15]
9	$[0, Y, -X]$	$\left[x+\frac{1}{2}, \frac{1}{2}, 0\right]$	[17,-20]
10	$[0, Y, X]$	$\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$	[18,-19]
11	$[Y, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}-x\right]$	[21,-24]
12	$[Y, -X, 0]$	$\left[\frac{1}{2}, 0, x+\frac{1}{2}\right]$	[22,-23]
13	$[0, -X, -Y]$	$\left[-x, \frac{1}{2}, 0\right]$	[25,-28]
14	$[0, X, -Y]$	$\left[x, \frac{1}{2}, 0\right]$	[26,-27]
15	$[-Y, 0, -X]$	$\left[0, -x, \frac{1}{2}\right]$	[29,-32]
16	$[-Y, 0, X]$	$\left[0, x, \frac{1}{2}\right]$	[30,-31]
17	$[-X, -Y, 0]$	$\left[\frac{1}{2}, 0, -x\right]$	[33,-36]
18	$[X, -Y, 0]$	$\left[\frac{1}{2}, 0, x\right]$	[34,-35]
19	$[-X, 0, Y]$	$\left[0, \frac{1}{2}-x, \frac{1}{2}\right]$	[37,-40]
20	$[X, 0, Y]$	$\left[0, x+\frac{1}{2}, \frac{1}{2}\right]$	[38,-39]
21	$[0, -Y, X]$	$\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$	[41,-44]
22	$[0, -Y, -X]$	$\left[x+\frac{1}{2}, \frac{1}{2}, 0\right]$	[42,-43]
23	$[-Y, -X, 0]$	$\left[\frac{1}{2}, 0, x+\frac{1}{2}\right]$	[45,-48]
24	$[-Y, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}-x\right]$	[46,-47]

Table 46: Wyckoff bond: 48e@12h, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, \frac{1}{2}, 0\right]$	[1]
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2}, 0\right]$	[2]
3	$[-X, Y, -Z]$	$\left[-x, \frac{1}{2}, 0\right]$	[3]

continued ...

Table 46

No.	vector	center	mapping
4	$[X, -Y, -Z]$	$\left[x, \frac{1}{2}, 0\right]$	[4]
5	$[Z, X, Y]$	$\left[0, x, \frac{1}{2}\right]$	[5]
6	$[Z, -X, -Y]$	$\left[0, -x, \frac{1}{2}\right]$	[6]
7	$[-Z, -X, Y]$	$\left[0, -x, \frac{1}{2}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[0, x, \frac{1}{2}\right]$	[8]
9	$[Y, Z, X]$	$\left[\frac{1}{2}, 0, x\right]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}, 0, -x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[\frac{1}{2}, 0, -x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}, 0, x\right]$	[12]
13	$[Y, X, -Z]$	$\left[0, x+\frac{1}{2}, \frac{1}{2}\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[0, \frac{1}{2}-x, \frac{1}{2}\right]$	[14]
15	$[Y, -X, Z]$	$\left[0, \frac{1}{2}-x, \frac{1}{2}\right]$	[15]
16	$[-Y, X, Z]$	$\left[0, x+\frac{1}{2}, \frac{1}{2}\right]$	[16]
17	$[X, Z, -Y]$	$\left[x+\frac{1}{2}, \frac{1}{2}, 0\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$	[19]
20	$[X, -Z, Y]$	$\left[x+\frac{1}{2}, \frac{1}{2}, 0\right]$	[20]
21	$[Z, Y, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}-x\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{1}{2}, 0, x+\frac{1}{2}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{2}, 0, x+\frac{1}{2}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}-x\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[-x, \frac{1}{2}, 0\right]$	[25]
26	$[X, Y, -Z]$	$\left[x, \frac{1}{2}, 0\right]$	[26]
27	$[X, -Y, Z]$	$\left[x, \frac{1}{2}, 0\right]$	[27]
28	$[-X, Y, Z]$	$\left[-x, \frac{1}{2}, 0\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[0, -x, \frac{1}{2}\right]$	[29]
30	$[-Z, X, Y]$	$\left[0, x, \frac{1}{2}\right]$	[30]
31	$[Z, X, -Y]$	$\left[0, x, \frac{1}{2}\right]$	[31]
32	$[Z, -X, Y]$	$\left[0, -x, \frac{1}{2}\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[\frac{1}{2}, 0, -x\right]$	[33]
34	$[Y, -Z, X]$	$\left[\frac{1}{2}, 0, x\right]$	[34]
35	$[-Y, Z, X]$	$\left[\frac{1}{2}, 0, x\right]$	[35]
36	$[Y, Z, -X]$	$\left[\frac{1}{2}, 0, -x\right]$	[36]
37	$[-Y, -X, Z]$	$\left[0, \frac{1}{2}-x, \frac{1}{2}\right]$	[37]
38	$[Y, X, Z]$	$\left[0, x+\frac{1}{2}, \frac{1}{2}\right]$	[38]
39	$[-Y, X, -Z]$	$\left[0, x+\frac{1}{2}, \frac{1}{2}\right]$	[39]
40	$[Y, -X, -Z]$	$\left[0, \frac{1}{2}-x, \frac{1}{2}\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$	[41]
42	$[X, -Z, -Y]$	$\left[x+\frac{1}{2}, \frac{1}{2}, 0\right]$	[42]
43	$[X, Z, Y]$	$\left[x+\frac{1}{2}, \frac{1}{2}, 0\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{1}{2}, 0, x+\frac{1}{2}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}-x\right]$	[46]
47	$[Z, -Y, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}-x\right]$	[47]
48	$[Z, Y, X]$	$\left[\frac{1}{2}, 0, x+\frac{1}{2}\right]$	[48]

* Wyckoff site: 16i, site symmetry: .3.

Table 47: Wyckoff bond: `16a@16i`, bond symmetry: `.3.`

No.	vector	center	mapping
1	$[X,\ X,\ X]$	$[x,\ x,\ x]$	`[1,5,9]`
2	$[-X,\ -X,\ X]$	$[-x,\ -x,\ x]$	`[2,7,12]`
3	$[-X,\ X,\ -X]$	$[-x,\ x,\ -x]$	`[3,8,10]`
4	$[X,\ -X,\ -X]$	$[x,\ -x,\ -x]$	`[4,6,11]`
5	$[X,\ X,\ -X]$	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[13,17,21]`
6	$[-X,\ -X,\ -X]$	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	`[14,19,24]`
7	$[X,\ -X,\ X]$	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	`[15,20,22]`
8	$[-X,\ X,\ X]$	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[16,18,23]`
9	$[-X,\ -X,\ -X]$	$[-x,\ -x,\ -x]$	`[25,29,33]`
10	$[X,\ X,\ -X]$	$[x,\ x,\ -x]$	`[26,31,36]`
11	$[X,\ -X,\ X]$	$[x,\ -x,\ x]$	`[27,32,34]`
12	$[-X,\ X,\ X]$	$[-x,\ x,\ x]$	`[28,30,35]`
13	$[-X,\ -X,\ X]$	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	`[37,41,45]`
14	$[X,\ X,\ X]$	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[38,43,48]`
15	$[-X,\ X,\ -X]$	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[39,44,46]`
16	$[X,\ -X,\ -X]$	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	`[40,42,47]`

Table 48: Wyckoff bond: `48b@16i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ x,\ x]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$[-x,\ -x,\ x]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$[-x,\ x,\ -x]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$[x,\ -x,\ -x]$	`[4]`
5	$[Z,\ X,\ Y]$	$[x,\ x,\ x]$	`[5]`
6	$[Z,\ -X,\ -Y]$	$[x,\ -x,\ -x]$	`[6]`
7	$[-Z,\ -X,\ Y]$	$[-x,\ -x,\ x]$	`[7]`
8	$[-Z,\ X,\ -Y]$	$[-x,\ x,\ -x]$	`[8]`
9	$[Y,\ Z,\ X]$	$[x,\ x,\ x]$	`[9]`
10	$[-Y,\ Z,\ -X]$	$[-x,\ x,\ -x]$	`[10]`
11	$[Y,\ -Z,\ -X]$	$[x,\ -x,\ -x]$	`[11]`
12	$[-Y,\ -Z,\ X]$	$[-x,\ -x,\ x]$	`[12]`
13	$[Y,\ X,\ -Z]$	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[13]`
14	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	`[14]`
15	$[Y,\ -X,\ Z]$	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	`[15]`
16	$[-Y,\ X,\ Z]$	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[16]`
17	$[X,\ Z,\ -Y]$	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[17]`
18	$[-X,\ Z,\ Y]$	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[18]`
19	$[-X,\ -Z,\ -Y]$	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	`[19]`
20	$[X,\ -Z,\ Y]$	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	`[20]`
21	$[Z,\ Y,\ -X]$	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[21]`
22	$[Z,\ -Y,\ X]$	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	`[22]`
23	$[-Z,\ Y,\ X]$	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[23]`
24	$[-Z,\ -Y,\ -X]$	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	`[24]`
25	$[-X,\ -Y,\ -Z]$	$[-x,\ -x,\ -x]$	`[25]`

continued ...

Table 48

No.	vector	center	mapping
26	$[X, Y, -Z]$	$[x, x, -x]$	[26]
27	$[X, -Y, Z]$	$[x, -x, x]$	[27]
28	$[-X, Y, Z]$	$[-x, x, x]$	[28]
29	$[-Z, -X, -Y]$	$[-x, -x, -x]$	[29]
30	$[-Z, X, Y]$	$[-x, x, x]$	[30]
31	$[Z, X, -Y]$	$[x, x, -x]$	[31]
32	$[Z, -X, Y]$	$[x, -x, x]$	[32]
33	$[-Y, -Z, -X]$	$[-x, -x, -x]$	[33]
34	$[Y, -Z, X]$	$[x, -x, x]$	[34]
35	$[-Y, Z, X]$	$[-x, x, x]$	[35]
36	$[Y, Z, -X]$	$[x, x, -x]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{2} - x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	[37]
38	$[Y, X, Z]$	$\left[x + \frac{1}{2}, x + \frac{1}{2}, x + \frac{1}{2}\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	[39]
40	$[Y, -X, -Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, \frac{1}{2} - x\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{2} - x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, \frac{1}{2} - x\right]$	[42]
43	$[X, Z, Y]$	$\left[x + \frac{1}{2}, x + \frac{1}{2}, x + \frac{1}{2}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{1}{2} - x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, \frac{1}{2} - x\right]$	[46]
47	$[Z, -Y, -X]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, \frac{1}{2} - x\right]$	[47]
48	$[Z, Y, X]$	$\left[x + \frac{1}{2}, x + \frac{1}{2}, x + \frac{1}{2}\right]$	[48]

* Wyckoff site: 24j, site symmetry: ..2

Table 49: Wyckoff bond: 24a@24j, bond symmetry: ..2

No.	vector	center	mapping
1	$[Y, X, -X]$	$\left[\frac{1}{4}, y, y + \frac{1}{2}\right]$	[1,-18]
2	$[-Y, -X, -X]$	$\left[\frac{3}{4}, -y, y + \frac{1}{2}\right]$	[2,-20]
3	$[-Y, X, X]$	$\left[\frac{3}{4}, y, \frac{1}{2} - y\right]$	[3,-17]
4	$[Y, -X, X]$	$\left[\frac{1}{4}, -y, \frac{1}{2} - y\right]$	[4,-19]
5	$[-X, Y, X]$	$\left[y + \frac{1}{2}, \frac{1}{4}, y\right]$	[5,-15]
6	$[-X, -Y, -X]$	$\left[y + \frac{1}{2}, \frac{3}{4}, -y\right]$	[6,-13]
7	$[X, -Y, X]$	$\left[\frac{1}{2} - y, \frac{3}{4}, y\right]$	[7,-16]
8	$[X, Y, -X]$	$\left[\frac{1}{2} - y, \frac{1}{4}, -y\right]$	[8,-14]
9	$[X, -X, Y]$	$\left[y, y + \frac{1}{2}, \frac{1}{4}\right]$	[9,-21]
10	$[-X, -X, -Y]$	$\left[-y, y + \frac{1}{2}, \frac{3}{4}\right]$	[10,-23]
11	$[X, X, -Y]$	$\left[y, \frac{1}{2} - y, \frac{3}{4}\right]$	[11,-22]
12	$[-X, X, Y]$	$\left[-y, \frac{1}{2} - y, \frac{1}{4}\right]$	[12,-24]
13	$[-Y, -X, X]$	$\left[\frac{3}{4}, -y, \frac{1}{2} - y\right]$	[25,-42]
14	$[Y, X, X]$	$\left[\frac{1}{4}, y, \frac{1}{2} - y\right]$	[26,-44]
15	$[Y, -X, -X]$	$\left[\frac{1}{4}, -y, y + \frac{1}{2}\right]$	[27,-41]
16	$[-Y, X, -X]$	$\left[\frac{3}{4}, y, y + \frac{1}{2}\right]$	[28,-43]
17	$[X, -Y, -X]$	$\left[\frac{1}{2} - y, \frac{3}{4}, -y\right]$	[29,-39]
18	$[X, Y, X]$	$\left[\frac{1}{2} - y, \frac{1}{4}, y\right]$	[30,-37]

continued ...

Table 49

No.	vector	center	mapping
19	$[-X, Y, -X]$	$\left[y+\frac{1}{2}, \frac{1}{4}, -y\right]$	[31,-40]
20	$[-X, -Y, X]$	$\left[y+\frac{1}{2}, \frac{3}{4}, y\right]$	[32,-38]
21	$[-X, X, -Y]$	$\left[-y, \frac{1}{2}-y, \frac{3}{4}\right]$	[33,-45]
22	$[X, X, Y]$	$\left[y, \frac{1}{2}-y, \frac{1}{4}\right]$	[34,-47]
23	$[-X, -X, Y]$	$\left[-y, y+\frac{1}{2}, \frac{1}{4}\right]$	[35,-46]
24	$[X, -X, -Y]$	$\left[y, y+\frac{1}{2}, \frac{3}{4}\right]$	[36,-48]

Table 50: Wyckoff bond: 24b@24j, bond symmetry: ..2

No.	vector	center	mapping
1	$[0, X, X]$	$\left[\frac{1}{4}, y, y+\frac{1}{2}\right]$	[1,18]
2	$[0, -X, X]$	$\left[\frac{3}{4}, -y, y+\frac{1}{2}\right]$	[2,20]
3	$[0, X, -X]$	$\left[\frac{3}{4}, y, \frac{1}{2}-y\right]$	[3,17]
4	$[0, -X, -X]$	$\left[\frac{1}{4}, -y, \frac{1}{2}-y\right]$	[4,19]
5	$[X, 0, X]$	$\left[y+\frac{1}{2}, \frac{1}{4}, y\right]$	[5,15]
6	$[X, 0, -X]$	$\left[y+\frac{1}{2}, \frac{3}{4}, -y\right]$	[6,13]
7	$[-X, 0, X]$	$\left[\frac{1}{2}-y, \frac{3}{4}, y\right]$	[7,16]
8	$[-X, 0, -X]$	$\left[\frac{1}{2}-y, \frac{1}{4}, -y\right]$	[8,14]
9	$[X, X, 0]$	$\left[y, y+\frac{1}{2}, \frac{1}{4}\right]$	[9,21]
10	$[-X, X, 0]$	$\left[-y, y+\frac{1}{2}, \frac{3}{4}\right]$	[10,23]
11	$[X, -X, 0]$	$\left[y, \frac{1}{2}-y, \frac{3}{4}\right]$	[11,22]
12	$[-X, -X, 0]$	$\left[-y, \frac{1}{2}-y, \frac{1}{4}\right]$	[12,24]
13	$[0, -X, -X]$	$\left[\frac{3}{4}, -y, \frac{1}{2}-y\right]$	[25,42]
14	$[0, X, -X]$	$\left[\frac{1}{4}, y, \frac{1}{2}-y\right]$	[26,44]
15	$[0, -X, X]$	$\left[\frac{1}{4}, -y, y+\frac{1}{2}\right]$	[27,41]
16	$[0, X, X]$	$\left[\frac{3}{4}, y, y+\frac{1}{2}\right]$	[28,43]
17	$[-X, 0, -X]$	$\left[\frac{1}{2}-y, \frac{3}{4}, -y\right]$	[29,39]
18	$[-X, 0, X]$	$\left[\frac{1}{2}-y, \frac{1}{4}, y\right]$	[30,37]
19	$[X, 0, -X]$	$\left[y+\frac{1}{2}, \frac{1}{4}, -y\right]$	[31,40]
20	$[X, 0, X]$	$\left[y+\frac{1}{2}, \frac{3}{4}, y\right]$	[32,38]
21	$[-X, -X, 0]$	$\left[-y, \frac{1}{2}-y, \frac{3}{4}\right]$	[33,45]
22	$[X, -X, 0]$	$\left[y, \frac{1}{2}-y, \frac{1}{4}\right]$	[34,47]
23	$[-X, X, 0]$	$\left[-y, y+\frac{1}{2}, \frac{1}{4}\right]$	[35,46]
24	$[X, X, 0]$	$\left[y, y+\frac{1}{2}, \frac{3}{4}\right]$	[36,48]

Table 51: Wyckoff bond: 48c@24j, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, y, y+\frac{1}{2}\right]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{3}{4}, -y, y+\frac{1}{2}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{3}{4}, y, \frac{1}{2}-y\right]$	[3]
4	$[X, -Y, -Z]$	$\left[\frac{1}{4}, -y, \frac{1}{2}-y\right]$	[4]
5	$[Z, X, Y]$	$\left[y+\frac{1}{2}, \frac{1}{4}, y\right]$	[5]
6	$[Z, -X, -Y]$	$\left[y+\frac{1}{2}, \frac{3}{4}, -y\right]$	[6]

continued ...

Table 51

No.	vector	center	mapping
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}-y, \frac{3}{4}, y\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{1}{2}-y, \frac{1}{4}, -y\right]$	[8]
9	$[Y, Z, X]$	$\left[y, y+\frac{1}{2}, \frac{1}{4}\right]$	[9]
10	$[-Y, Z, -X]$	$\left[-y, y+\frac{1}{2}, \frac{3}{4}\right]$	[10]
11	$[Y, -Z, -X]$	$\left[y, \frac{1}{2}-y, \frac{3}{4}\right]$	[11]
12	$[-Y, -Z, X]$	$\left[-y, \frac{1}{2}-y, \frac{1}{4}\right]$	[12]
13	$[Y, X, -Z]$	$\left[y+\frac{1}{2}, \frac{3}{4}, -y\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{1}{2}-y, \frac{1}{4}, -y\right]$	[14]
15	$[Y, -X, Z]$	$\left[y+\frac{1}{2}, \frac{1}{4}, y\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{1}{2}-y, \frac{3}{4}, y\right]$	[16]
17	$[X, Z, -Y]$	$\left[\frac{3}{4}, y, \frac{1}{2}-y\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{4}, y, y+\frac{1}{2}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{4}, -y, \frac{1}{2}-y\right]$	[19]
20	$[X, -Z, Y]$	$\left[\frac{3}{4}, -y, y+\frac{1}{2}\right]$	[20]
21	$[Z, Y, -X]$	$\left[y, y+\frac{1}{2}, \frac{1}{4}\right]$	[21]
22	$[Z, -Y, X]$	$\left[y, \frac{1}{2}-y, \frac{3}{4}\right]$	[22]
23	$[-Z, Y, X]$	$\left[-y, y+\frac{1}{2}, \frac{3}{4}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[-y, \frac{1}{2}-y, \frac{1}{4}\right]$	[24]
25	$[-X, -Y, -Z]$	$\left[\frac{3}{4}, -y, \frac{1}{2}-y\right]$	[25]
26	$[X, Y, -Z]$	$\left[\frac{1}{4}, y, \frac{1}{2}-y\right]$	[26]
27	$[X, -Y, Z]$	$\left[\frac{1}{4}, -y, y+\frac{1}{2}\right]$	[27]
28	$[-X, Y, Z]$	$\left[\frac{3}{4}, y, y+\frac{1}{2}\right]$	[28]
29	$[-Z, -X, -Y]$	$\left[\frac{1}{2}-y, \frac{3}{4}, -y\right]$	[29]
30	$[-Z, X, Y]$	$\left[\frac{1}{2}-y, \frac{1}{4}, y\right]$	[30]
31	$[Z, X, -Y]$	$\left[y+\frac{1}{2}, \frac{1}{4}, -y\right]$	[31]
32	$[Z, -X, Y]$	$\left[y+\frac{1}{2}, \frac{3}{4}, y\right]$	[32]
33	$[-Y, -Z, -X]$	$\left[-y, \frac{1}{2}-y, \frac{3}{4}\right]$	[33]
34	$[Y, -Z, X]$	$\left[y, \frac{1}{2}-y, \frac{1}{4}\right]$	[34]
35	$[-Y, Z, X]$	$\left[-y, y+\frac{1}{2}, \frac{1}{4}\right]$	[35]
36	$[Y, Z, -X]$	$\left[y, y+\frac{1}{2}, \frac{3}{4}\right]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{2}-y, \frac{1}{4}, y\right]$	[37]
38	$[Y, X, Z]$	$\left[y+\frac{1}{2}, \frac{3}{4}, y\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{2}-y, \frac{3}{4}, -y\right]$	[39]
40	$[Y, -X, -Z]$	$\left[y+\frac{1}{2}, \frac{1}{4}, -y\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{4}, -y, y+\frac{1}{2}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[\frac{3}{4}, -y, \frac{1}{2}-y\right]$	[42]
43	$[X, Z, Y]$	$\left[\frac{3}{4}, y, y+\frac{1}{2}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{4}, y, \frac{1}{2}-y\right]$	[44]
45	$[-Z, -Y, X]$	$\left[-y, \frac{1}{2}-y, \frac{3}{4}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[-y, y+\frac{1}{2}, \frac{1}{4}\right]$	[46]
47	$[Z, -Y, -X]$	$\left[y, \frac{1}{2}-y, \frac{1}{4}\right]$	[47]
48	$[Z, Y, X]$	$\left[y, y+\frac{1}{2}, \frac{3}{4}\right]$	[48]

* Wyckoff site: 24k, site symmetry: m..

Table 52: Wyckoff bond: `24a@24k`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[0,\ X,\ Y]$	$[0,\ y,\ z]$	`[1,28]`
2	$[0,\ -X,\ Y]$	$[0,\ -y,\ z]$	`[2,27]`
3	$[0,\ X,\ -Y]$	$[0,\ y,\ -z]$	`[3,26]`
4	$[0,\ -X,\ -Y]$	$[0,\ -y,\ -z]$	`[4,25]`
5	$[Y,\ 0,\ X]$	$[z,\ 0,\ y]$	`[5,32]`
6	$[Y,\ 0,\ -X]$	$[z,\ 0,\ -y]$	`[6,31]`
7	$[-Y,\ 0,\ X]$	$[-z,\ 0,\ y]$	`[7,30]`
8	$[-Y,\ 0,\ -X]$	$[-z,\ 0,\ -y]$	`[8,29]`
9	$[X,\ Y,\ 0]$	$[y,\ z,\ 0]$	`[9,36]`
10	$[-X,\ Y,\ 0]$	$[-y,\ z,\ 0]$	`[10,35]`
11	$[X,\ -Y,\ 0]$	$[y,\ -z,\ 0]$	`[11,34]`
12	$[-X,\ -Y,\ 0]$	$[-y,\ -z,\ 0]$	`[12,33]`
13	$[X,\ 0,\ -Y]$	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[13,40]`
14	$[-X,\ 0,\ -Y]$	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[14,39]`
15	$[X,\ 0,\ Y]$	$\left[y+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[15,38]`
16	$[-X,\ 0,\ Y]$	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[16,37]`
17	$[0,\ Y,\ -X]$	$\left[\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$	`[17,44]`
18	$[0,\ Y,\ X]$	$\left[\frac{1}{2},\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$	`[18,43]`
19	$[0,\ -Y,\ -X]$	$\left[\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$	`[19,42]`
20	$[0,\ -Y,\ X]$	$\left[\frac{1}{2},\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$	`[20,41]`
21	$[Y,\ X,\ 0]$	$\left[z+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[21,48]`
22	$[Y,\ -X,\ 0]$	$\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[22,47]`
23	$[-Y,\ X,\ 0]$	$\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[23,46]`
24	$[-Y,\ -X,\ 0]$	$\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[24,45]`

Table 53: Wyckoff bond: `24b@24k`, bond symmetry: `m..`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$[0,\ y,\ z]$	`[1,-28]`
2	$[-X,\ 0,\ 0]$	$[0,\ -y,\ z]$	`[2,-27]`
3	$[-X,\ 0,\ 0]$	$[0,\ y,\ -z]$	`[3,-26]`
4	$[X,\ 0,\ 0]$	$[0,\ -y,\ -z]$	`[4,-25]`
5	$[0,\ X,\ 0]$	$[z,\ 0,\ y]$	`[5,-32]`
6	$[0,\ -X,\ 0]$	$[z,\ 0,\ -y]$	`[6,-31]`
7	$[0,\ -X,\ 0]$	$[-z,\ 0,\ y]$	`[7,-30]`
8	$[0,\ X,\ 0]$	$[-z,\ 0,\ -y]$	`[8,-29]`
9	$[0,\ 0,\ X]$	$[y,\ z,\ 0]$	`[9,-36]`
10	$[0,\ 0,\ -X]$	$[-y,\ z,\ 0]$	`[10,-35]`
11	$[0,\ 0,\ -X]$	$[y,\ -z,\ 0]$	`[11,-34]`
12	$[0,\ 0,\ X]$	$[-y,\ -z,\ 0]$	`[12,-33]`
13	$[0,\ X,\ 0]$	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[13,-40]`
14	$[0,\ -X,\ 0]$	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[14,-39]`
15	$[0,\ -X,\ 0]$	$\left[y+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[15,-38]`
16	$[0,\ X,\ 0]$	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[16,-37]`
17	$[X,\ 0,\ 0]$	$\left[\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$	`[17,-44]`

continued ...

Table 53

No.	vector	center	mapping
18	$[-X, 0, 0]$	$\left[\frac{1}{2}, z+\frac{1}{2}, y+\frac{1}{2}\right]$	[18,-43]
19	$[-X, 0, 0]$	$\left[\frac{1}{2}, \frac{1}{2}-z, \frac{1}{2}-y\right]$	[19,-42]
20	$[X, 0, 0]$	$\left[\frac{1}{2}, \frac{1}{2}-z, y+\frac{1}{2}\right]$	[20,-41]
21	$[0, 0, -X]$	$\left[z+\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}\right]$	[21,-48]
22	$[0, 0, X]$	$\left[z+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}\right]$	[22,-47]
23	$[0, 0, X]$	$\left[\frac{1}{2}-z, y+\frac{1}{2}, \frac{1}{2}\right]$	[23,-46]
24	$[0, 0, -X]$	$\left[\frac{1}{2}-z, \frac{1}{2}-y, \frac{1}{2}\right]$	[24,-45]

Table 54: Wyckoff bond: 48c@24k, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, y, z]$	[1]
2	$[-X, -Y, Z]$	$[0, -y, z]$	[2]
3	$[-X, Y, -Z]$	$[0, y, -z]$	[3]
4	$[X, -Y, -Z]$	$[0, -y, -z]$	[4]
5	$[Z, X, Y]$	$[z, 0, y]$	[5]
6	$[Z, -X, -Y]$	$[z, 0, -y]$	[6]
7	$[-Z, -X, Y]$	$[-z, 0, y]$	[7]
8	$[-Z, X, -Y]$	$[-z, 0, -y]$	[8]
9	$[Y, Z, X]$	$[y, z, 0]$	[9]
10	$[-Y, Z, -X]$	$[-y, z, 0]$	[10]
11	$[Y, -Z, -X]$	$[y, -z, 0]$	[11]
12	$[-Y, -Z, X]$	$[-y, -z, 0]$	[12]
13	$[Y, X, -Z]$	$\left[y+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-z\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{1}{2}-y, \frac{1}{2}, \frac{1}{2}-z\right]$	[14]
15	$[Y, -X, Z]$	$\left[y+\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{1}{2}-y, \frac{1}{2}, z+\frac{1}{2}\right]$	[16]
17	$[X, Z, -Y]$	$\left[\frac{1}{2}, z+\frac{1}{2}, \frac{1}{2}-y\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{2}, z+\frac{1}{2}, y+\frac{1}{2}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}-z, \frac{1}{2}-y\right]$	[19]
20	$[X, -Z, Y]$	$\left[\frac{1}{2}, \frac{1}{2}-z, y+\frac{1}{2}\right]$	[20]
21	$[Z, Y, -X]$	$\left[z+\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}\right]$	[21]
22	$[Z, -Y, X]$	$\left[z+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{2}-z, y+\frac{1}{2}, \frac{1}{2}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{1}{2}-z, \frac{1}{2}-y, \frac{1}{2}\right]$	[24]
25	$[-X, -Y, -Z]$	$[0, -y, -z]$	[25]
26	$[X, Y, -Z]$	$[0, y, -z]$	[26]
27	$[X, -Y, Z]$	$[0, -y, z]$	[27]
28	$[-X, Y, Z]$	$[0, y, z]$	[28]
29	$[-Z, -X, -Y]$	$[-z, 0, -y]$	[29]
30	$[-Z, X, Y]$	$[-z, 0, y]$	[30]
31	$[Z, X, -Y]$	$[z, 0, -y]$	[31]
32	$[Z, -X, Y]$	$[z, 0, y]$	[32]
33	$[-Y, -Z, -X]$	$[-y, -z, 0]$	[33]
34	$[Y, -Z, X]$	$[y, -z, 0]$	[34]

continued ...

Table 54

No.	vector	center	mapping
35	$[-Y, Z, X]$	$[-y, z, 0]$	[35]
36	$[Y, Z, -X]$	$[y, z, 0]$	[36]
37	$[-Y, -X, Z]$	$[\frac{1}{2} - y, \frac{1}{2}, z + \frac{1}{2}]$	[37]
38	$[Y, X, Z]$	$[y + \frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}]$	[38]
39	$[-Y, X, -Z]$	$[\frac{1}{2} - y, \frac{1}{2}, \frac{1}{2} - z]$	[39]
40	$[Y, -X, -Z]$	$[y + \frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z]$	[40]
41	$[-X, -Z, Y]$	$[\frac{1}{2}, \frac{1}{2} - z, y + \frac{1}{2}]$	[41]
42	$[X, -Z, -Y]$	$[\frac{1}{2}, \frac{1}{2} - z, \frac{1}{2} - y]$	[42]
43	$[X, Z, Y]$	$[\frac{1}{2}, z + \frac{1}{2}, y + \frac{1}{2}]$	[43]
44	$[-X, Z, -Y]$	$[\frac{1}{2}, z + \frac{1}{2}, \frac{1}{2} - y]$	[44]
45	$[-Z, -Y, X]$	$[\frac{1}{2} - z, \frac{1}{2} - y, \frac{1}{2}]$	[45]
46	$[-Z, Y, -X]$	$[\frac{1}{2} - z, y + \frac{1}{2}, \frac{1}{2}]$	[46]
47	$[Z, -Y, -X]$	$[z + \frac{1}{2}, \frac{1}{2} - y, \frac{1}{2}]$	[47]
48	$[Z, Y, X]$	$[z + \frac{1}{2}, y + \frac{1}{2}, \frac{1}{2}]$	[48]

* Wyckoff site: 481, site symmetry: 1

Table 55: Wyckoff bond: 48a@481, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	[1]
2	$[-X, -Y, Z]$	$[-x, -y, z]$	[2]
3	$[-X, Y, -Z]$	$[-x, y, -z]$	[3]
4	$[X, -Y, -Z]$	$[x, -y, -z]$	[4]
5	$[Z, X, Y]$	$[z, x, y]$	[5]
6	$[Z, -X, -Y]$	$[z, -x, -y]$	[6]
7	$[-Z, -X, Y]$	$[-z, -x, y]$	[7]
8	$[-Z, X, -Y]$	$[-z, x, -y]$	[8]
9	$[Y, Z, X]$	$[y, z, x]$	[9]
10	$[-Y, Z, -X]$	$[-y, z, -x]$	[10]
11	$[Y, -Z, -X]$	$[y, -z, -x]$	[11]
12	$[-Y, -Z, X]$	$[-y, -z, x]$	[12]
13	$[Y, X, -Z]$	$[y + \frac{1}{2}, x + \frac{1}{2}, \frac{1}{2} - z]$	[13]
14	$[-Y, -X, -Z]$	$[\frac{1}{2} - y, \frac{1}{2} - x, \frac{1}{2} - z]$	[14]
15	$[Y, -X, Z]$	$[y + \frac{1}{2}, \frac{1}{2} - x, z + \frac{1}{2}]$	[15]
16	$[-Y, X, Z]$	$[\frac{1}{2} - y, x + \frac{1}{2}, z + \frac{1}{2}]$	[16]
17	$[X, Z, -Y]$	$[x + \frac{1}{2}, z + \frac{1}{2}, \frac{1}{2} - y]$	[17]
18	$[-X, Z, Y]$	$[\frac{1}{2} - x, z + \frac{1}{2}, y + \frac{1}{2}]$	[18]
19	$[-X, -Z, -Y]$	$[\frac{1}{2} - x, \frac{1}{2} - z, \frac{1}{2} - y]$	[19]
20	$[X, -Z, Y]$	$[x + \frac{1}{2}, \frac{1}{2} - z, y + \frac{1}{2}]$	[20]
21	$[Z, Y, -X]$	$[z + \frac{1}{2}, y + \frac{1}{2}, \frac{1}{2} - x]$	[21]
22	$[Z, -Y, X]$	$[z + \frac{1}{2}, \frac{1}{2} - y, x + \frac{1}{2}]$	[22]
23	$[-Z, Y, X]$	$[\frac{1}{2} - z, y + \frac{1}{2}, x + \frac{1}{2}]$	[23]
24	$[-Z, -Y, -X]$	$[\frac{1}{2} - z, \frac{1}{2} - y, \frac{1}{2} - x]$	[24]
25	$[-X, -Y, -Z]$	$[-x, -y, -z]$	[25]
26	$[X, Y, -Z]$	$[x, y, -z]$	[26]
27	$[X, -Y, Z]$	$[x, -y, z]$	[27]

continued ...

Table 55

No.	vector	center	mapping
28	$[-X, Y, Z]$	$[-x, y, z]$	[28]
29	$[-Z, -X, -Y]$	$[-z, -x, -y]$	[29]
30	$[-Z, X, Y]$	$[-z, x, y]$	[30]
31	$[Z, X, -Y]$	$[z, x, -y]$	[31]
32	$[Z, -X, Y]$	$[z, -x, y]$	[32]
33	$[-Y, -Z, -X]$	$[-y, -z, -x]$	[33]
34	$[Y, -Z, X]$	$[y, -z, x]$	[34]
35	$[-Y, Z, X]$	$[-y, z, x]$	[35]
36	$[Y, Z, -X]$	$[y, z, -x]$	[36]
37	$[-Y, -X, Z]$	$\left[\frac{1}{2} - y, \frac{1}{2} - x, z + \frac{1}{2}\right]$	[37]
38	$[Y, X, Z]$	$\left[y + \frac{1}{2}, x + \frac{1}{2}, z + \frac{1}{2}\right]$	[38]
39	$[-Y, X, -Z]$	$\left[\frac{1}{2} - y, x + \frac{1}{2}, \frac{1}{2} - z\right]$	[39]
40	$[Y, -X, -Z]$	$\left[y + \frac{1}{2}, \frac{1}{2} - x, \frac{1}{2} - z\right]$	[40]
41	$[-X, -Z, Y]$	$\left[\frac{1}{2} - x, \frac{1}{2} - z, y + \frac{1}{2}\right]$	[41]
42	$[X, -Z, -Y]$	$\left[x + \frac{1}{2}, \frac{1}{2} - z, \frac{1}{2} - y\right]$	[42]
43	$[X, Z, Y]$	$\left[x + \frac{1}{2}, z + \frac{1}{2}, y + \frac{1}{2}\right]$	[43]
44	$[-X, Z, -Y]$	$\left[\frac{1}{2} - x, z + \frac{1}{2}, \frac{1}{2} - y\right]$	[44]
45	$[-Z, -Y, X]$	$\left[\frac{1}{2} - z, \frac{1}{2} - y, x + \frac{1}{2}\right]$	[45]
46	$[-Z, Y, -X]$	$\left[\frac{1}{2} - z, y + \frac{1}{2}, \frac{1}{2} - x\right]$	[46]
47	$[Z, -Y, -X]$	$\left[z + \frac{1}{2}, \frac{1}{2} - y, \frac{1}{2} - x\right]$	[47]
48	$[Z, Y, X]$	$\left[z + \frac{1}{2}, y + \frac{1}{2}, x + \frac{1}{2}\right]$	[48]